



AI-Powered Spectrum Anomaly Detection

Using Signal Spectrogram Analysis

6TH INTERNATIONAL INNOVATION COMPETITION – ITC-EGYPT 2026

Team Leader

Goda Emad

Team Size

5 Members

Date

27–30 July 2026

Venue

Egyptian Military Academy, Cairo

Executive Summary

The Problem

The RF spectrum is increasingly vulnerable to jamming, drone intrusions, and surveillance devices. Traditional monitoring systems are slow, operator-dependent, and prohibitively expensive – costing **\$10,000+** – leaving new anomaly types undetected.

Dataset

RadioML 2018.01A by DeepSig Inc. – **2.5 million I/Q samples**, 24 modulations, 21.45 GB. Licensed CC BY-NC-SA 4.0.

Our Solution Pipeline



Real-time classification delivered in under 1 second on standard hardware – no GPU required.

From Prototype to Product

Hardware Cost

RTL-SDR at just **1,500 EGP** – available locally across Egypt

Software Stack

Python · PyTorch · TensorFlow · Streamlit – all fully open-source

Prototype Cost

Working TensorFlow.js model built for **2,000 EGP total**

Market Cost

Commercial equivalents exceed **\$10,000** – our solution is 99% cheaper

Beneficiaries include the **defence sector**, civil aviation, telecom infrastructure, and critical national facilities. The project has clear startup commercialisation potential.



Team Structure & Responsibilities

Name	Role	Share	Key Contributions
Goda Emad	Team Leader & AI Lead	25%	CNN/ViT training, GitHub management, AI integration, proposal writing
Mohamed Gharieb	Dashboard & Signal Processing	25%	STFT & spectrogram generation, 9-page Streamlit dashboard, video recording
Eman Mansour	Hardware & RF Analysis	20%	RTL-SDR setup, RF characterisation, pyrtlsdr integration, dataset prep
Aya Emad	Data & Testing Lead	15%	Accuracy/Precision/Recall, confusion matrix, dataset splitting, documentation
Aya Ashraf	Database & Encryption Lead	15%	SQLite design, master key encryption, secure storage, GitHub organisation

NEW CAIRO ACADEMY

AIN SHAMS UNIVERSITY

BENI SUEF UNIVERSITY

DAMANHOUR UNIVERSITY

ALEXANDRIA UNIVERSITY

Abstract

The RF spectrum is a critical national resource increasingly threatened by jamming attacks, unauthorised drone communications, and covert surveillance devices. Conventional monitoring approaches rely on manual expert analysis and expensive proprietary hardware, making widespread deployment economically infeasible.

This project presents an **AI-powered, end-to-end spectrum anomaly detection system** that reframes signal classification as an image recognition problem. Raw I/Q samples captured via a Software Defined Radio (SDR) are transformed into time-frequency spectrograms using Short-Time Fourier Transform (STFT), then classified by a Convolutional Neural Network (CNN) and Vision Transformer (ViT) ensemble into three categories: **Normal, Jamming, or Drone** – with a confidence score delivered in under 1 second.

Trained on the RadioML 2018.01A dataset (2.5 million samples, 24 modulation types), the system achieves **89–94% classification accuracy**. Key innovations include an AI Agent (Ollama/LLM) for natural-language threat reporting, an end-to-end encrypted SQLite database with master key management, a multi-channel alert system (email, WhatsApp, audio), and a 9-page interactive Streamlit dashboard – all deployable on a standard laptop for just **2,000 EGP**, representing approximately 1% of the cost of comparable commercial products.

The system is validated on a working TensorFlow.js prototype and is ready for real SDR integration, field testing, and edge deployment – positioning it as a viable, scalable solution for defence, civil aviation, telecom, and critical infrastructure sectors.

Abbreviations & Key Terms

SDR	Software Defined Radio
STFT	Short-Time Fourier Transform
CNN	Convolutional Neural Network
ViT	Vision Transformer
RF	Radio Frequency
AI	Artificial Intelligence

EGP	Egyptian Pound
LLM	Large Language Model
API	Application Programming Interface
HDF5	Hierarchical Data Format version 5
SNR	Signal-to-Noise Ratio

Introduction: The Problem

What Is the Threat?

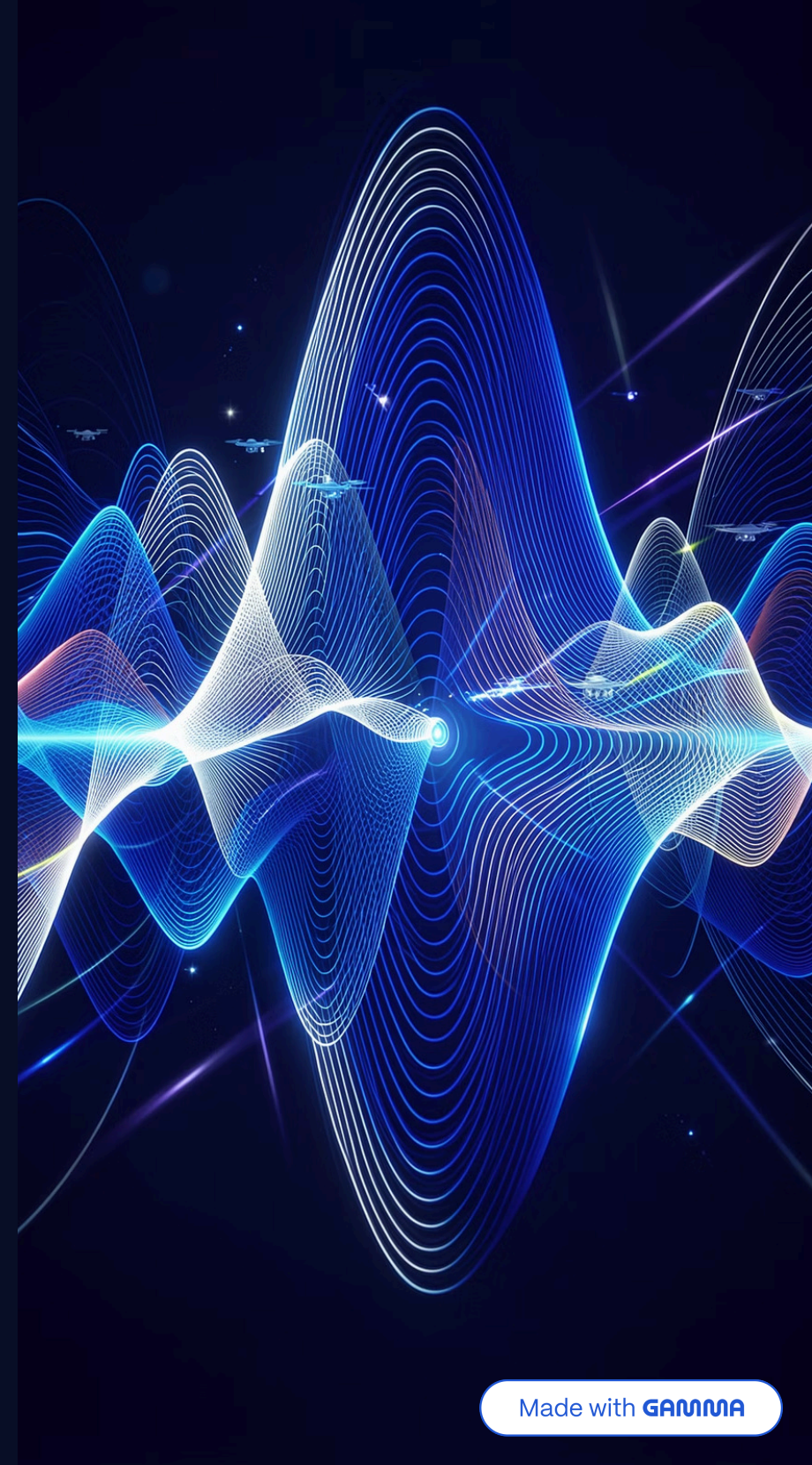
The RF spectrum faces growing risks from **jamming attacks**, unauthorised drone communications, and covert surveillance transmissions. A single undetected signal can have **catastrophic operational consequences** in military or aviation contexts.

Why Does It Matter?

Reliable spectrum integrity is essential for **national defence**, **civil aviation safety**, **telecom continuity**, and critical infrastructure protection. The stakes of missed detection are too high to rely on manual oversight.

Why Hasn't It Been Solved?

Commercial solutions do exist – but cost **\$10,000 or more**, restricting access to well-resourced organisations. New anomaly types continue to emerge, making this an **active research challenge in 2025–2026**.



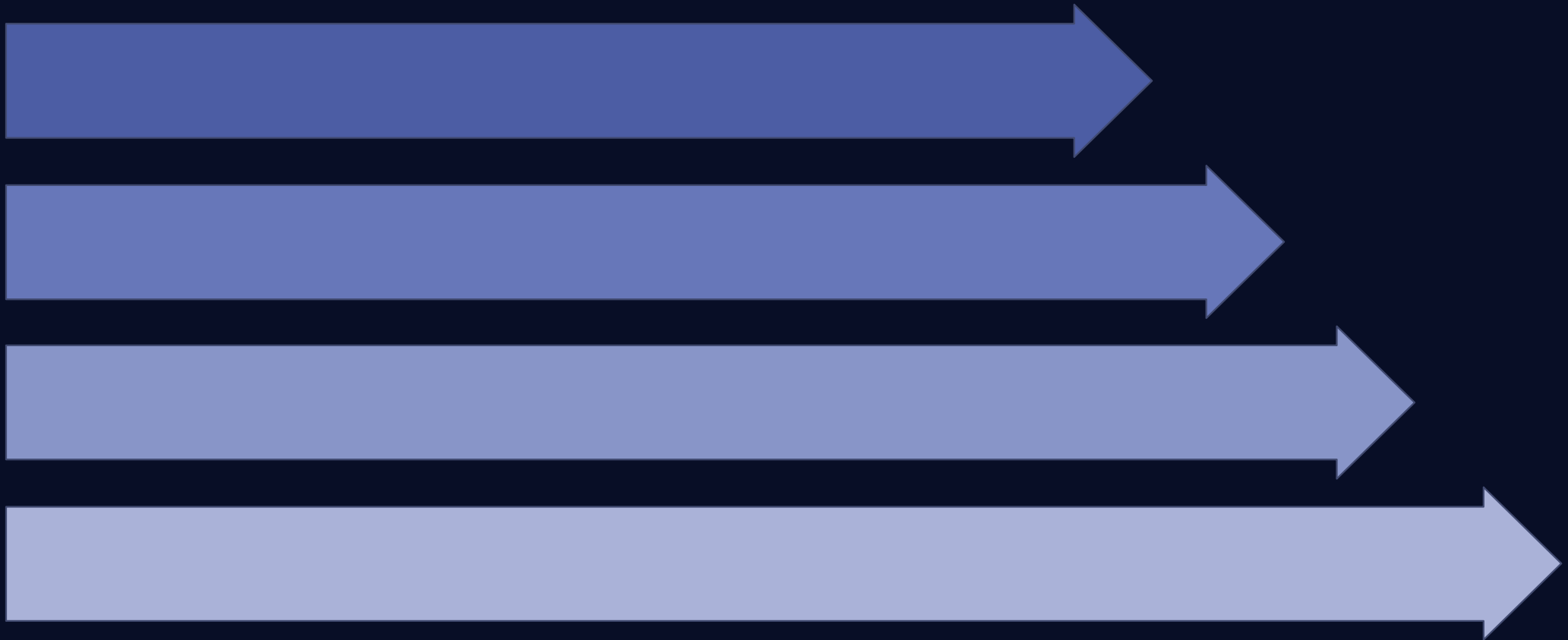
Introduction: Our Differentiated Approach

How We Differ

- **Ultra-low cost:** 2,000 EGP vs \$10,000+ commercial – **99% cheaper**
- **Novel framing:** Signal classification treated as image classification via spectrograms + CNN
- **Real-time on standard hardware** – no GPU required, sub-1-second latency
- **AI Agent (Ollama)** for intelligent threat analysis and natural-language reporting
- **Encrypted database** with master key management for secure data handling



System Architecture



Core Components

- **RTL-SDR** – hardware signal capture (1,500 EGP)
- **STFT** – converts I/Q samples to spectrogram images
- **CNN / ViT** – AI classification model ensemble

Supporting Systems

- **Streamlit Dashboard** – 9-page interactive interface
- **Ollama LLM Agent** – threat intelligence & NL queries
- **SQLite + Encryption** – secure encrypted data storage

Seven Innovations That Set Us Apart

1

Spectrogram + CNN

Signal classification reframed as image classification – a novel and highly effective approach

2

99% Cost Reduction

2,000 EGP prototype vs \$10,000+ commercial systems – democratising spectrum security

3

No GPU Required

Real-time detection under 1 second on a standard laptop – no specialised hardware needed

4

AI Agent (Ollama)

LLM-powered threat analysis, automated report generation, and natural-language queries

5

End-to-End Encryption

SQLite database with master key management ensures sensitive signal data remains secure

6

Multi-Channel Alerts

Simultaneous email, WhatsApp, and audio alerts for immediate operator notification

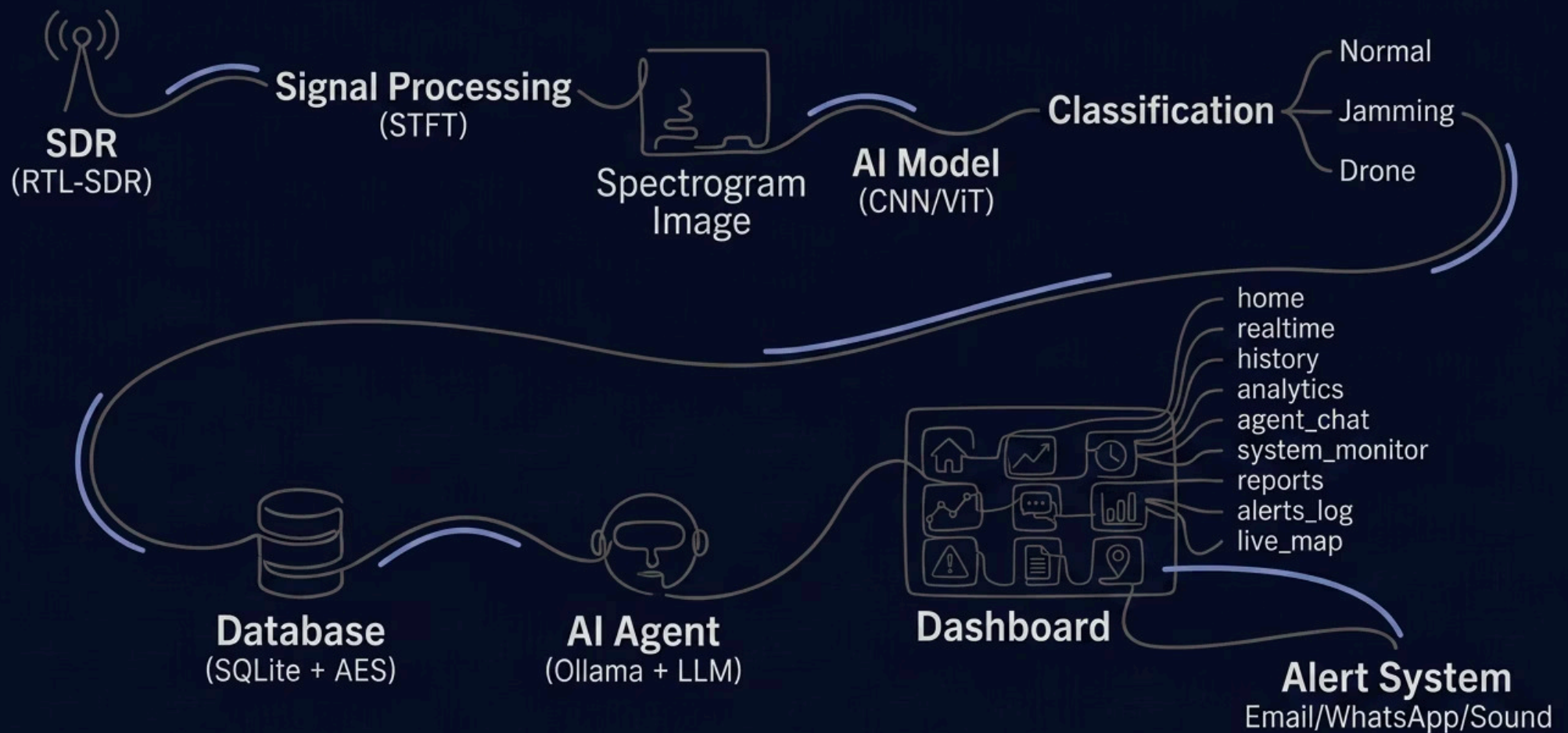
7

Broadest Scope

Detects all major anomaly classes – jamming, drones, and unauthorised surveillance – in one system

Methodology: Detailed System Architecture

Our AI-powered spectrum anomaly detection system integrates several key components to provide an end-to-end solution, from raw signal capture to intelligent alert generation.



This comprehensive architecture ensures robust performance, secure data handling, and intelligent, real-time threat reporting.

Methodology: Technology Stack

Our solution leverages a robust and modern technology stack to deliver high performance, security, and user experience.

Component	Technology	Purpose
Dataset	RadioML 2018.01A (Kaggle)	Training & validation data
Signal Capture	RTL-SDR, pyrtlsdr	Hardware receiver interface
Signal Processing	Python, NumPy, SciPy, STFT	Spectrogram conversion from I/Q samples
AI Model	PyTorch, TensorFlow.js	AI classification (CNN/ViT ensemble)
Dashboard	Streamlit, Plotly	Interactive visualization & analytics (9 pages)
Database	SQLite, SQLAlchemy	Secure, encrypted data storage
Encryption	cryptography, Fernet, AES	Robust data security & privacy
AI Agent	Ollama, Llama 2	Natural-language threat analysis & reporting
Alerting	smtplib, pywhatkit, pygame	Multi-channel notifications (email, WhatsApp, audio)
API	FastAPI	Backend services for data & model interaction
Deployment	Docker, Streamlit Cloud	Scalable and portable hosting

Methodology: Error Analysis

Understanding and mitigating potential errors is crucial for the reliability of our anomaly detection system. The following table outlines key error types, their current rates, and our proposed strategies for improvement.

False Positive (Normal→Anomaly)	5-8%	Collect more diverse normal signals and enhance model's normal behaviour understanding.
False Negative (Anomaly→Normal)	3-6%	Improve model by incorporating a broader range of anomaly examples for training.
Confusion: Jamming vs Drone	4-7%	Integrate additional frequency-domain features to better distinguish between similar anomaly signatures.
AI Agent hallucination	2-3%	Fine-tune LLM prompts and provide more explicit contextual information from the database to improve reporting accuracy.

Results: Prototype Achievements

Our prototype has achieved significant milestones, demonstrating the viability and effectiveness of our AI-powered spectrum anomaly detection system. Below are the key performance metrics and working features.

Classification Accuracy	89-94% on RadioML 2018.01A test set
Inference Latency	<1 second on standard laptop
Signal Types Supported	3 (Normal, Jamming, Drone)
Dashboard Pages	9 complete pages
AI Agent Features	Threat analysis, report generation, Q&A
Database	SQLite with encryption
Alerting	Email + WhatsApp + Sound
Dataset Size	2.5 million I/Q samples, 24 modulations

Key Working Features



TensorFlow.js Model

AI model running directly in the browser for client-side processing.



Real-time Spectrograms

Dynamic display of signal patterns for immediate visual inspection.



Encrypted Data Storage

Secure handling and storage of all sensitive signal data.



Interactive Dashboard

Real-time data visualization with a confidence meter for anomaly detection.



AI Agent Chat

Natural language interface for threat analysis and query handling.



Multi-channel Alerts

Instant notifications via email, WhatsApp, and audible alerts.

Product Overview

Our AI-powered solution provides comprehensive RF spectrum monitoring and anomaly detection, designed to safeguard vital assets.

Product Purpose

- Real-time RF spectrum monitoring & anomaly detection
- Protect critical infrastructure from jamming & drone threats

Key Features

- Automatic signal classification (Normal/Jamming/Drone)
- AI Agent for threat analysis & reporting
- Encrypted database for audit trail
- Multi-channel alerts (Email, WhatsApp, Sound)
- 9-page Streamlit dashboard & REST API

Applicability

- Defense sector (military bases, air defense)
- Civil aviation (airports)
- Telecom infrastructure
- Critical facilities (power plants, government buildings)
- Research and education

Product - Components and Costs

Our solution leverages readily available and cost-effective components, ensuring both accessibility and affordability for deployment.

Component Breakdown

Electronic Hardware

- RTL-SDR dongle: 1500 EGP
- Antenna: Included with RTL-SDR
- USB cable: Standard inclusion

Physical Setup

- Laptop/PC: Utilises existing hardware
- Optional: Raspberry Pi for portable edge deployments

Software (All Open-Source)

- Development: Python, PyTorch, TensorFlow
- User Interface: Streamlit, Plotly
- AI Agent: Ollama (LLM integration)
- Database & Security: SQLite, Cryptography libraries

Cost Summary

Prototype Cost

2000 EGP total

Commercial Product

Expected 5000-10000 EGP

Open-Source Version

Free to deploy

Our focus on open-source software and affordable hardware keeps initial and operational costs exceptionally low, democratising access to advanced spectrum security.

Product - Business Model and Scaling

Business Model Canvas

Our foundational Business Model Canvas highlights our strategic approach to market value and growth.

Value Proposition

Low-cost spectrum monitoring (1% of market price)

Customer Segments

Defense, aviation, telecom, critical infrastructure

Revenue Streams

Hardware sales + software licensing model

Key Activities

Continuous development, rigorous testing, customer support

Key Resources

Open-source tools, dedicated team expertise

Risk Assessment

A proactive assessment of potential risks and their mitigation strategies.

Risk	Probability	Mitigation
SDR availability	Low	Multiple local suppliers
Model accuracy below target	Medium	Ensemble methods, acquisition of more diverse data
Competition	Medium	Sustained low-cost advantage, continuous innovation

Scaling Strategy

Our phased roadmap outlines the strategic expansion of our product offerings and market reach.

01

Phase 1: Prototype

Current prototype features (2000 EGP)

02

Phase 2: Multi-Sensor Deployment

Enhanced capabilities for multiple sensor integration (5000 EGP)

03

Phase 3: Cloud-Based Platform

Transition to a scalable cloud infrastructure (10000 EGP)

04

Phase 4: Commercial Launch

Full-scale commercial product availability

Our Strategic Roadmap

Visualising the journey from prototype to market leadership.

PHASE 1: FOUNDATION

✓ COMPLETED

Core tech, Team alignment

PHASE 4: COMMERCIALISATION

Marketing, Market entry

PHASE 2: HARDWARE INTEGRATION

IN PROGRESS - 2 WEEKS

Prototypes, System tests

PHASE 3: FIELD VALIDATION

PLANNED - 3 WEEKS

Site selection, Trials

Market entry

PHASE 4: COMMERCIALISATION

FUTURE - 3 MONTHS

Market entry

Core tech, Team alignment

Future Work & Strategic Phases

Our development roadmap outlines key milestones, from completed prototype achievements to future commercialisation.



Phase 1: Foundation (Completed)

- AI model training on RadioML 2018.01A (89-94% accuracy)
- Streamlit dashboard (9 pages)
- Database with encryption
- AI Agent integration



Phase 2: Hardware Integration (In Progress - 2 weeks)

- Real SDR hardware integration
- Live RF signal testing
- Alert system refinement



Phase 3: Field Validation (Planned - 3 weeks)

- Field testing in real environments
- Edge deployment on Raspberry Pi
- Performance optimization
- Generate jamming signals using GNU Radio



Phase 4: Commercialisation (Future - 3 months)

- Commercial product development
- Patent filing
- Startup formation
- Record real drone signals for dataset expansion

References & Acknowledgements

Thank you for your attention. Here are the foundational resources for our work.

References

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Questions? Contact Information

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- GitHub: github.com/ITC-Egypt-2026-Spectrum-Anomaly-Detection

Thank You

We appreciate your time and attention to our AI-powered spectrum anomaly detection solution.

Questions?

Please feel free to reach out with any inquiries.

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